

ARANSAS COUNTY AP, TX, USA

WMO: 722524

Lat: 28.084N

Lon: 97.046W

Elev: 7

StdP: 101.24

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 12972

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.1	3.8	-6.0	2.3	8.3	-3.4	2.9	9.2	11.6	10.5	10.7	10.5	5.3	0	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.0	34.0	25.9	33.3	26.1	32.8	26.1	27.5	31.5	27.2	31.2	26.9	30.8	5.3	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.4	21.9	29.0	26.2	21.7	29.0	26.0	21.4	29.0	87.1	31.7	85.7	31.0	84.2	30.9	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.0	8.8	8.2	-0.6	36.6	1.9	2.0	-2.0	38.0	-3.1	39.1	-4.2	40.2	-5.6	41.6		
			-2.2	28.1	2.0	0.6	-3.7	28.5	-4.9	28.8	-6.0	29.1	-7.5	29.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.6	13.9	16.0	19.0	22.6	26.0	28.8	29.3	29.7	27.9	24.1	18.9	14.9	
	DBStd	6.64	4.95	4.82	4.19	3.17	2.17	1.40	1.19	1.16	2.22	3.85	4.70	4.92	
	HDD10.0	48	21	9	3	0	0	0	0	0	0	0	3	12	
	HDD18.3	478	150	91	43	7	0	0	0	0	0	7	51	130	
	CDD10.0	4657	143	178	283	378	497	563	597	610	537	438	269	163	
	CDD18.3	2045	14	26	65	136	239	313	338	352	287	187	67	23	
	CDH23.3	20924	32	46	171	674	2097	3756	4273	4529	3274	1690	330	53	
	CDH26.7	8112	4	4	17	104	527	1545	1897	2159	1351	468	34	1	
Wind	WSAvg	4.5	4.2	4.5	4.9	5.4	5.3	4.8	4.3	4.1	4.0	4.2	4.1	4.1	
Precipitation	PrecAvg	845	53	52	64	58	90	75	72	69	134	89	59	47	
	PrecMax	1225	171	159	183	220	293	247	455	218	397	297	238	158	
	PrecMin	406	0	0	0	7	0	0	0	7	28	0	0	12	
	PrecStd	241	42	42	47	51	79	58	107	49	86	73	60	37	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.3	26.6	28.4	31.3	32.3	34.2	34.0	35.2	35.2	32.5	28.8	26.3	
		MCWB	18.3	19.6	19.6	19.8	24.3	24.8	26.1	25.9	25.0	25.3	23.8	21.5	
	2%	DB	23.8	24.6	26.4	28.9	31.2	33.0	33.3	34.0	33.4	31.3	27.7	24.8	
		MCWB	19.3	20.1	21.3	22.3	24.7	26.0	26.2	26.3	25.7	25.4	22.8	20.8	
	5%	DB	22.5	23.2	25.2	27.7	30.2	32.4	32.8	33.4	32.6	30.2	26.6	23.5	
		MCWB	18.7	19.5	20.5	22.4	24.6	26.0	26.2	26.3	25.8	24.7	21.9	20.3	
	10%	DB	21.1	22.1	23.9	26.8	29.2	32.0	32.3	32.8	32.0	29.1	25.3	22.3	
		MCWB	18.3	19.1	20.2	22.1	24.3	26.0	26.2	26.3	25.7	24.0	21.7	19.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.6	22.1	23.1	25.0	26.8	28.0	27.6	28.0	27.9	27.3	24.9	23.0
			MCDB	23.5	24.5	26.2	28.1	30.0	31.4	31.6	32.5	31.7	30.2	27.3	25.2
2%		WB	20.7	21.3	22.4	24.3	26.1	27.3	27.2	27.5	27.2	26.5	24.0	22.2	
		MCDB	22.3	23.3	25.0	27.2	29.1	30.6	31.3	32.0	31.0	29.3	26.5	23.9	
5%		WB	19.9	20.7	21.8	23.8	25.6	26.9	26.9	27.2	26.8	25.8	23.3	21.3	
		MCDB	21.4	22.3	23.9	26.4	28.7	30.4	31.0	31.6	30.6	28.7	25.6	22.9	
10%		WB	19.0	20.0	21.3	23.2	25.2	26.6	26.7	26.9	26.4	25.3	22.6	20.3	
		MCDB	20.5	21.4	23.2	25.6	28.4	30.2	30.8	31.1	30.2	28.4	24.8	22.0	
Mean Daily Temperature Range	5% DB	MDBR	8.9	8.1	7.7	7.2	6.4	6.4	6.3	7.0	7.5	8.2	8.9	9.0	
		MCDBR	9.7	8.3	7.4	7.5	7.0	6.9	6.7	7.5	8.0	7.3	8.3	9.1	
		MCWBR	6.5	5.2	3.9	3.7	2.9	2.4	2.3	2.5	3.0	3.5	4.7	6.4	
	5% WB	MCDBR	8.1	7.0	6.2	5.9	5.6	5.9	6.1	6.7	6.8	6.5	7.4	8.1	
		MCWBR	6.3	5.2	3.7	3.1	2.6	2.5	2.4	2.5	3.0	3.8	5.2	6.6	
Clear-Sky Solar Irradiance	taub		0.359	0.367	0.386	0.432	0.476	0.451	0.469	0.450	0.441	0.393	0.379	0.348	
	taud		2.484	2.466	2.409	2.287	2.196	2.330	2.295	2.349	2.338	2.466	2.453	2.534	
	Ebn at Noon		882	904	905	868	827	842	826	841	840	866	852	879	
	Edh at Noon		94	103	116	135	148	129	133	125	123	102	96	86	
All-Sky Solar Radiation	RadAvg		2.97	3.50	4.54	5.40	6.08	6.70	6.58	6.29	5.21	4.53	3.40	2.68	
	RadStd		0.33	0.58	0.46	0.51	0.42	0.36	0.57	0.34	0.45	0.47	0.36	0.35	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	-0.28	N/A	N/A	N/A	N/A	
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+85	

Nomenclature: See separate page